



Dissipative particle dynamics simulation of cell entry into a micro-channel

Lüwen Zhou^{a,*}, Shiliang Feng^a, Hantao Liu^b, Jianzhong Chang^b^a Faculty of Mechanical Engineering and Mechanics, Ningbo University, Ningbo, Zhejiang 315211, China^b The Laboratory of Energy & Environment Engineering and Computational Fluid Dynamics, North University of China, Taiyuan 030051, China

ARTICLE INFO

MSC:
00-01
99-00Keywords:
Cell entry
Micro-channel
Dissipative particle dynamics (DPD)

ABSTRACT

Cell deformability is an important biomarker which can be used to distinguish and sort between healthy and cancer cells. In this paper, we presented a dissipative particle dynamics (DPD) model for investigating cell entry into micro-channels. The cell membrane is represented by a network of DPD particles (beads) connected by worm-like chain (WLC) springs, which is able to mimic the viscoelastic effect of the membrane. The entry process of benign breast epithelial cells (MCF-10A) and non-metastatic tumor breast cells (MCF-7) through a constricted micro-channel are comparatively investigated using this DPD model. It is shown that both the time histories of the cell displacement and the dynamic behaviors of cell entry agree with experimental observations. The entry time of MCF-10A cell is approximately four times of that of MCF-7 cell since MCF-10A cells are stiffer than MCF-7 cells. It is demonstrated that the presented DPD method is effective in modeling cell deformability, and the obtained results can be helpful in understanding how cells with different mechanical properties respond to physical loads.

1. Introduction

Dynamical behaviors of migration and deformation variations of cells in confined environment are probably caused by pathological changes in mechanical properties of cells, which may be closely related to several cell diseases. These variations are often facilitated by the altering in the mechanical behaviors of cells such as large changes of elastic modulus [1]. Modern physiology and medicine have established the relationship of mechanical variations between healthy and pathological cells. For example, compared to healthy cells, diseased cells such as cancer cells are known to have different stiffness and elasticity [2]. Such differences can be used to distinguish between normal and diseased cells [3,4]. Recently, increased micro-fluidic devices were designed and applied in medical diagnostics to diagnose and treat cells disease [5,6]. Thus, investigating cell entry into micro channels can be fundamentally important and it is of significance to understand how cells with different mechanical properties respond to physical loads, and to further design more effective micro-fluidic devices.

Continuum cell models, treat the cell as comprising materials with certain continuum material properties, are main approaches to model the mechanical dynamics of cells. Generally, continuum models can be classified into solid models and liquid drop models. Practice shows that solid models may fail for large cell deformations [7]. Liquid drop models treat the cell as a liquid drop, which can be used to model large cell deformation. The Newtonian liquid drop model was developed by Yeung and Evans [8], and it can describe large cell deformations well

only when the progress is slow [9]. In order to consider the effects of the nucleus on cell deformation, the compound drop model was proposed [10], which assumed the nucleus to be an encapsulated liquid drop. Recently, Fong et al. [11] presented a modified compound drop model, which is able to account for stiffness, elasticity, and viscosity of both the cortex and the nucleus, to model breast cancer cell entry into a constricted micro-channel.

Although continuum cell models are easier and more straightforward to use in computing the mechanical properties of the cells [12], it provides less insights into the detailed molecular mechanical events. For this reason, more accurate micro and nanostructural models were developed later. Boey et al. [13] and Li et al. [14] established and developed the spectrin-level RBC model. It was successfully validated against experimental data of the mechanical response of an individual cell. However it involves limited degrees of freedom and application of the model in flow simulations requires prohibitively expensive computations. For this reason, Pivkin and Karniadakis [15] originally developed a coarse-grained model based on the spectrin-level RBC model. A more systematic and rigorous procedure to derive coarse-grained RBC models was present by Fedosov et al. [16]. The RBC is modeled by DPD particles and captures the elastic response at both small and large deformations. Xiao et al. [17] used this model to simulate the motion of two RBCs with intercellular interactions in a stenosed microvessel. In addition, Fedosov et al. [18] and Ye et al. [19,20] also extended this model to model malaria-infected RBCs. Ye et al. [21] also extended this model to study the relationships between the transit time and the

* Corresponding author.

E-mail addresses: zhoulvwen@nbn.edu.cn (L. Zhou), changjz01@sina.com (J. Chang).

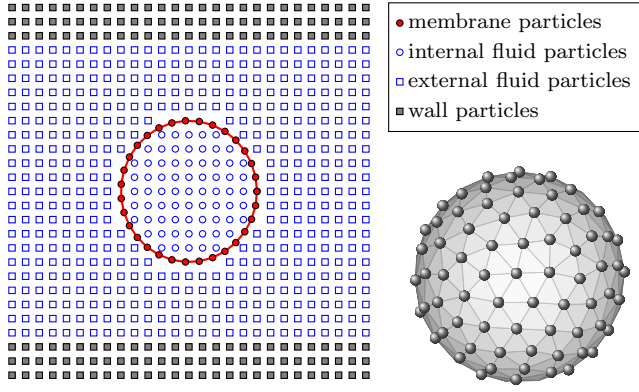


Fig. 1. DPD system with four types of particles. Cell membrane model is represented by a network of springs linked DPD particles.

mechanical properties of a cell flowing through a microchannel with a single constriction. Furthermore, they [22] investigate this microfluidic chip design and examine its potential in performances. Li et al. [23] used a two-component protein-scale RBC model to simulate the traversal of the interendothelial slit in the human spleen.

In this work, we present a DPD model for simulating the entry process of MCF-10A and MCF-7 cells through a constricted micro-channel. Cell model is described in Section 2. Numerical simulations of MCF-10A and MCF-7 cells through a constricted micro-channel are presented and analyzed in Sections 3 and 4. The paper concludes in Section 5 with some remarks.

2. Methods

2.1. DPD method

Dissipative particle dynamics (DPD) [24] is a mesoscopic method which describes clusters of molecules (particle) moving together in a Lagrangian fashion subject to soft quadratic potentials. The total force acting on a DPD particle i is given by a sum over all particles j that lie within a fixed cut-off distance, of three pairwise-additive forces:

$$\mathbf{f}_i = \sum_{j \neq i} \mathbf{F}_{ij}^C + \mathbf{F}_{ij}^D + \mathbf{F}_{ij}^R \quad (1)$$

where $\mathbf{F}_{ij}^C$ is a conservative force, $\mathbf{F}_{ij}^D$ and $\mathbf{F}_{ij}^R$ is a dissipative force and a random force. The conservative force acts to give particle a chemical identity, while the dissipative and random forces together form a thermostat that keeps the mean temperature of the system constant. For more detailed information about methodology and applications of DPD, see Refs. [24–26]. In our DPD model, all parameters used are the same as those in the [27] except for the cut-off radius being adjusted to 1.5 times unit length.

2.2. Cell membrane model

In this section, we construct cell membrane model using DPD. There are four types of particles in our DPD system, as shown in Fig. 1. The non-slip boundary condition is implemented by using frozen particles in the solid obstacle areas together with the Maxwellian reflection model when a fluid particle enters a thin reflecting boundary layer. This solid boundary treatment has been proven to be effective in preventing unphysical particle penetration and density oscillations can be controlled in a reasonably low level [26,28–30]. The cell membrane structure is defined by a two-dimensional triangular network on the spherical surface. Each link of triangular network is modelled by nonlinear WLC spring model [31]. The force between membrane particles includes the elastic and viscous parts. The elastic part is characterized by an energy

potential, given by

$$U(\{\mathbf{r}_i\}) = U_{\text{in-plane}} + U_{\text{bending}} + U_{\text{volume}} \quad (2)$$

The in-plane energy term $U_{\text{in-plane}}$ indicates the elastic energy stored in the membrane, which is defined as [31]

$$U_{\text{in-plane}} = \sum_{i \in 1 \dots N_s} \left(\frac{k_B T l_i^{\max}}{4p_i} \frac{3x_i^2 - 2x_i^3}{1 - x_i} + \frac{k_i}{x_i} \right) \quad (3)$$

where N_s is the number of chains; $x_i = l_i/l_i^{\max} \in (0, 1)$, l_i and $l_i^{\max}$ are the current and maximum length of spring i ; p_i is the persistence length; and k_i is the power force coefficient. The bending energy term U_{bending} in equation (2), which represents the bending resistance of the lipid bilayer, is assumed as

$$U_{\text{bending}} = \sum_{i \in 1 \dots N_s} k_b [1 - \cos(\theta_i - \theta_i^0)] \quad (4)$$

where k_b is the bending constant; θ_i is the instantaneous angle between two adjacent triangles having a common edge, and θ_i^0 is the spontaneous angle. The volume restraint term U_{volume} in equation (2), which account for incompressibility of the inner cytosol, is given by

$$U_{\text{volume}} = \frac{k_v (V_{\text{tot}} - V_{\text{tot}}^0)^2}{2V_{\text{tot}}^0} \quad (5)$$

where k_v is the volume restraint constant; V_{tot} and V_{tot}^0 are the current and desired global volumes of the cell.

The dissipative part in DPD model is insufficient to represent membrane viscosity. For this reason, we follow the general framework of the fluid particle model [32,33] to define an additional dissipative force $\mathbf{F}_{ij}^{D,\text{vis}}$ and random force $\mathbf{F}_{ij}^{R,\text{vis}}$ for each spring as

$$\mathbf{F}_{ij}^{D,\text{vis}} = -\gamma^T \mathbf{v}_{ij} - \gamma^C (\mathbf{v}_{ij} \cdot \hat{\mathbf{r}}_{ij}) \hat{\mathbf{r}}_{ij} \quad (6)$$

$$\mathbf{F}_{ij}^{R,\text{vis}} = \sqrt{2k_B T} \left(\sqrt{2\gamma^T} d\mathbf{W}_{ij}^S + \frac{1}{3} \sqrt{3\gamma^C - \gamma^T} \text{tr}[d\mathbf{W}_{ij}] \mathbf{I} \right) \hat{\mathbf{r}}_{ij} \quad (7)$$

where γ^T and γ^C are dissipative parameters, $\mathbf{v}_{ij}$ is the relative velocity of two beads of one spring. $\mathbf{I}$ is the unit second-order tensor, $\text{tr}[d\mathbf{W}_{ij}]$ is the trace of a random matrix of independent Wiener increments $d\mathbf{W}_{ij}$, and $d\mathbf{W}_{ij}^S = \mathbf{W}_{ij}^S - \text{tr}[d\mathbf{W}_{ij}] \mathbf{I}/3$ is the traceless symmetric part.

2.3. Parameters determination

In order to connect the above-described cell model to the macroscopic properties of cell, a relationship between macroscopic elastic properties of the network and model parameters has to be derived. Linear analysis [15,34] for a regular hexagonal network was applied to obtain its linear macroscopic properties with respect to the selected network parameters. The membrane shear modulus is given by [16]

$$\mu_0 = \frac{\sqrt{3}k_B T}{4p_i l_{i,0}} \left(\frac{3x_0 - 1}{4(1 - x_0)^3} + \frac{1}{4} \right) + \frac{2\sqrt{3}k_p}{4l_{i,0}^3} \quad (8)$$

where $x_0 = l_{j,0}/l_j^{\max}$, and the subscript ‘0’ refers to the stress-free state. The corresponding linear-elastic area-compression moduli K and Young’s moduli Y are given as

$$K = 2\mu_0, \quad Y = \frac{4K\mu_0}{K + \mu_0} \quad (9)$$

Equating the macroscopic bending energy of the Helfrich model [35] and the bending energy of the network model U_{bending} , yields that the macroscopic bending rigidity can be expressed in terms of the bending coefficient k_b as [16]

$$E_B = \frac{2}{\sqrt{3}} k_b \quad (10)$$

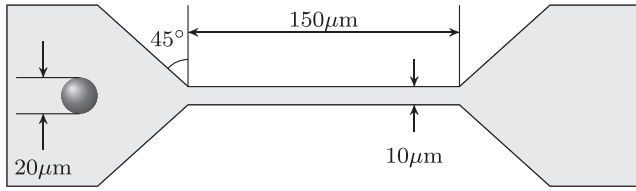


Fig. 2. Schematic of a micro-channel with a tapered entrance.

To simplify the general framework of the fluid particle model, γ^T was set as $3\gamma^C$ in Eqs. (6) and (7). The viscosity of cell membrane is therefore dependent on the dissipative part of DPD model and the additional viscous part

$$\mu_m = \sqrt{3}\gamma^T + \sqrt{3}\gamma^C/4, \quad (11)$$

2.4. Scaling of model and physical units

DPD simulations are conventionally performed in non-dimensionalized units, based on the characteristic physical dimensions of the system. In present paper, the characteristic scale for length, force, energy unit and time [32] are

$$\lambda = \frac{D^P}{D^M} \quad (12)$$

$$\zeta = \frac{F^P}{F^M} = \frac{Y^P}{Y^M} \lambda, \quad (13)$$

$$\varepsilon = \frac{(k_B T)^P}{(k_B T)^M} = \frac{Y^P}{Y^M} \lambda^2, \quad (14)$$

$$\tau = \frac{D^P}{D^M} \frac{\eta^P}{\eta^M} \frac{Y^M}{Y^P} \quad (15)$$

where the subscript “M” and “P” indicate “model” and “physics”, respectively. D denote the cell diameter, F is force, Y denote Young’s modulus, $k_B T$ is the energy unit, and η is viscosity.

3. Simulation

In Hou et al.’s study [4], a microfabricated fluidic channel design consisting of a straight channel and two reservoirs, as shown in Fig. 2, was used to study the biorheological behaviour of MCF-10A cells and MCF-7 cells to develop a method to distinguish between non-malignant and malignant cells. The fluidic channel is 150 μm in length and has

a square cross section area of 10 by 10 μm with a 45° tapered entrance. Flow in the micro-channel is driven by a differential pressure $\Delta p = 490.5$ Pa established between both ends of the setup. Under experimental conditions at room temperature of 22–24 °C, the cells have to deform to pass through the channel, allowing quantification of several physical parameters such as entry time, transit velocity and elongation index. In the present study, we numerically investigate the entry of MCF-10A cells and MCF-7 cells into a micro-channel by using method described in previous sections. The cell membrane is composed of 3318 DPD particles. The diameters of the cells are based on mean experimental values of 20 ± 5 μm , and in this paper it is taken to be $D^P = 20$ μm and $D^M = 20$. For MCF-7 cells, the Young’s modulus is taken to be $Y^P = 0.02$ dyn/cm = 2×10^{-5} N/m referenced from the range of literature values (0.005–0.035 dyn/cm) [11] and $Y^M = 20$. MCF-7 cells were found to have an apparent Young’s modulus significantly lower (1.4–1.8 times) than that of their non-malignant (MCF-10A) counterparts [36]. For MCF-10A cells, the Young’s modulus is taken to be 1.6 times of their MCF-7 counterparts. The selected value of bending rigidity is $k_c = 6.93 \times 10^{-20}$ J referenced from the range of literature values ($4\text{--}9 \times 10^{-20}$ J) [37]. Room temperature is taken to be $T^P = 296.15$ K and $(k_B T)^M = 0.02$.

It is not convenient for DPD simulation driven by a differential pressure as in the experiment. we use the following formula to translate differential pressure to a body force exerting on every DPD particle:

$$g^M = \frac{\Delta P S^P}{\rho^M L^M S^M} \frac{1}{\zeta} = \frac{\Delta P \lambda^3}{\rho^M L^P} \frac{1}{\zeta}, \quad (16)$$

where ΔP is the pressure difference between the two ends of the channel. S and L is the cross-sectional area and length of the channel. ρ the fluid density. ζ is the characteristic scale of force, given by Eq. (13).

4. Results

In this section, MCF-10A cell is firstly studied by the DPD model and compared with the experiment, and then MCF-7 cell is also studied and compared with simulation results of MCF-10A cell.

4.1. MCF-10A passing through a contracted micro-channel

In DPD simulation, the cell is initially positioned at a distance of 25 from the channel entrance to give the cell initial transient period to reach steady velocity before entry. Fig. 3 shows snapshots of MCF-10A cell through the micro-channel, and Fig. 4 shows the dependence of the cell center displacement. Velocity of the cell can be obtained from

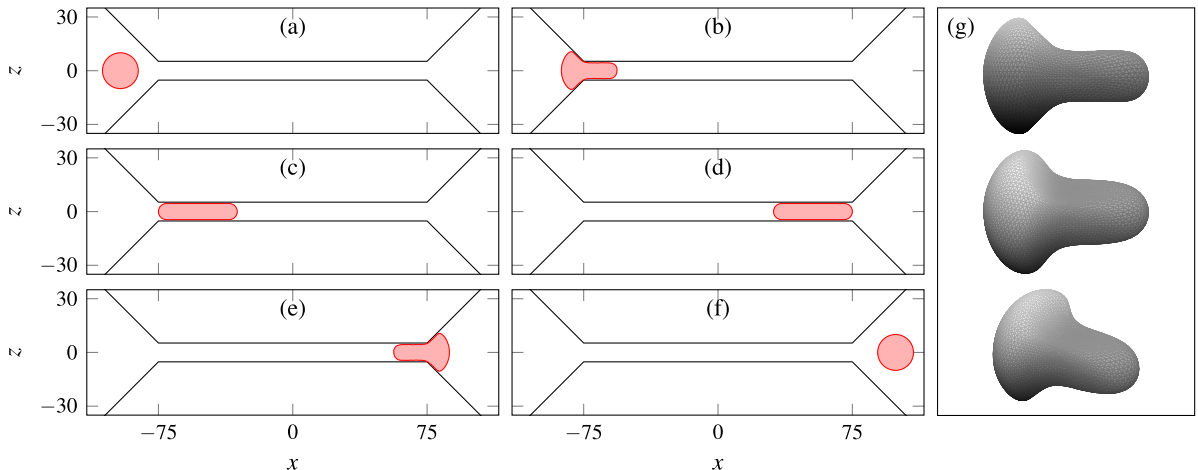


Fig. 3. Snapshots of MCF-10A cell traversing across a micro-channel from DPD simulation at $t = 0.00$ s (a), $t = 4.45$ s (b), $t = 16.41$ s (c), $t = 23.32$ s (d), $t = 24.92$ s (e) and $t = 29.79$ s (f), time is scaled by Eq. (15). (g) 3D shape of subfigure (b) at three angles of the view.

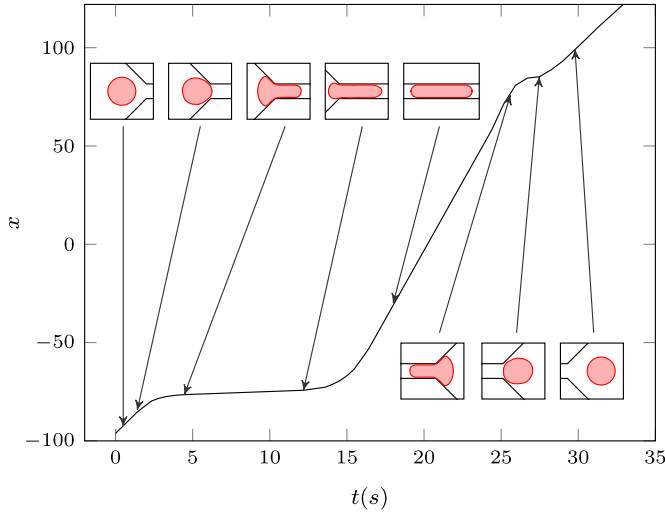


Fig. 4. Time-plots of MCF-10A cell center displacement from DPD simulation, time is scaled by Eq. (15).

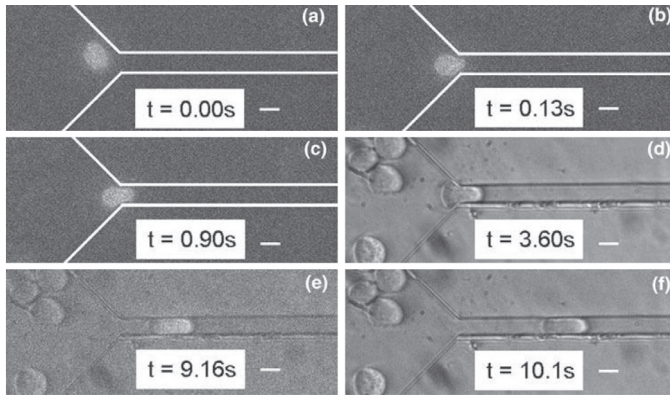


Fig. 5. Snapshots of a transfected MCF-10A breast epithelial cancer cell entering a micro-channel. The pressure gradient of the micro-channel is maintained at constant $p = 3.33 \text{ Pa}/\mu\text{m}$ (scale bar = $10 \mu\text{m}$) [11].

Fig. 4 since the velocity of the cell is the derivative of the displacement graph.

After a brief initial transient period, the cell velocity remains steady before touching the entrance, and we mark this moment as $t = 0.00\text{s}$ (see Fig. 3(a)). The cell touches the entrance and gradually deforms with reducing velocity after $t \approx 0.94\text{s}$. As shown in Fig. 3(b), midway through the entrance at $t \approx 4.45\text{s}$, the cell keeps changing its own conformation until its major axis reaches maximum, and the cytoplasmic velocity is observed to be uniform, which is very close to zero. As shown in Fig. 3(c), the cell fully enters the contracted micro-channel after $t \approx 16.41\text{s}$, and its velocity increases until a maximum is reached. Beyond the entry region, the cell deformation is minimal and its velocity keeps constant until its leading edge reaches the export (see Fig. 3(d)). As the front of the cell gradually leaves the micro-channel (see Fig. 3(e)), it restores its original structure with increasing velocity, and finally ends its travel in micro-channel (see Fig. 3(f)).

4.2. Comparisons with experiment

Leong et al. [11] using a high-speed camera recorded MCF-10A cell entry, as shown in Fig. 5. Fig. 3 shows snapshots of MCF-10A cell traversing across a micro-channel from DPD simulation. If we defined the cell entry time as the time interval between the leading edge and the trailing edge crossing the entrance of the micro-channel, we can see that the cell

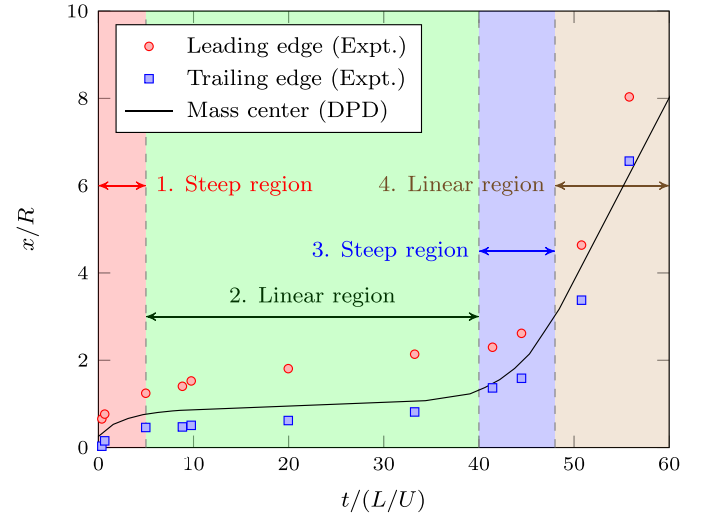


Fig. 6. Comparison of MCF-10A cell displacement with experimental data of Ref. [11], time is scaled by $t/(L/U)$.

entry time of our simulation is approximately 15 s, which is almost twice as much as that in experiment. There are many possible potential causes of the cell entry time of our simulation longer than that in experiment, such as uncertainty of some simulational parameters, and even the error from the experiment. But an important one may be volume restraint which the volume of the cell to be nearly constant in our simulations. However, cells seems to be compressible in experiment. The incompressibility of the cell in simulation may lead to more time required for the cell to enter the micro-channel, and the cell length stretch longer in the micro-channel.

Unfortunately, Leong et al. [11] only presented the process of cell entry instead of including the process of cell leaving the micro-channel. We cut out relevant portions and compare with the experimental data, as shown in Fig. 6. Fig. 6 shows the cell displacement for experimental data, as well as DPD model predictions. We observe good agreement between experimental and model results. The mass center of the cell predicted by our DPD model is between the leading edge and trailing edge by experimental observation. From Fig. 6, we can see that both profiles of DPD model and experimental data can be further divided into four distinct portions.

1. *Steep region*: The graph begins with a steep slope and this gradually decreases, as the cell is squeezed into the microchannel. This indicates that the cell's initial entry velocity into the channel is the highest but slows down as the cell attempts to squeeze into the micro-channel. The entry velocity slows down due to the increased resistance encountered by the cell. This initial non-linear portion lasts only for a short duration.
2. *Linear region*: This initial non-linear is followed immediately by a linear region with an almost horizontal slope. This linear portion of the graph is also the longest part in the entire entry-time region. This region is represented by the optical images in Fig. 5 (b), (d) which showed the main part of the cell squeezing into the channel.
3. *Steep region*: The slope increases again at the fellow region, corresponding to an increased entry velocity at the end of entry. This may be due to the decreased resistance encountered by the cell with the remaining cytoplasm deforming into the microchannel after the main body of the cell has entered the channel.
4. *Linear region*: This linear region again is at the final region after the cell has fully entered the channel. It happens when the cell deformation is minimal after the cell has fully deformed into the

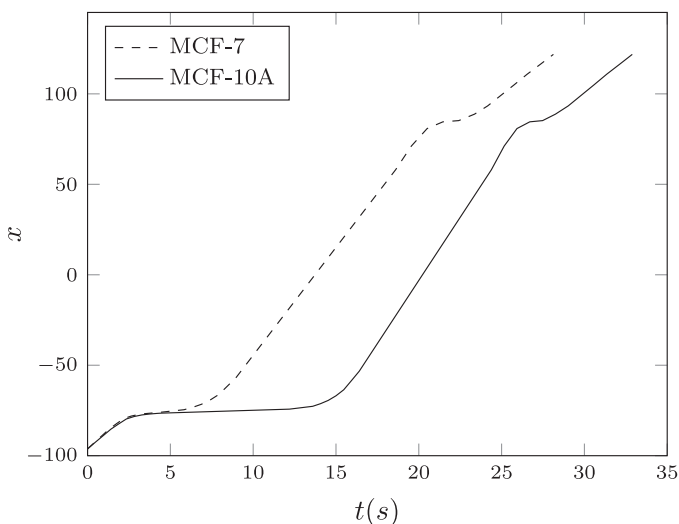


Fig. 7. Comparison of distance-to-origin profile of MCF-7 and MCF-10A, time is scaled by Eq. (15).

channel and its velocity keeps constant after its velocity increased to a maximum.

4.3. MCF-10A vs. MCF-7

Hou et al.'s [4] study demonstrated that the difference in entry times can be attributed to malignant MCF-7 cells being about 4–5 times less stiff compared to the approximately similar sizes benign MCF-10A cells. Fig. 7 shows the comparison of distance-to-origin profiles of MCF-7 and MCF-10A from DPD simulations. The general shape of the profiles of the two cells is similar. It will begin with a long entry region, followed by a steep linear transit region. The long entry region can also be further subdivided into four distinct portions as mentioned in the above section. As shown in Fig. 7, the entry region for MCF-10A is much longer, and this means that the cell takes a longer time to deform completely into the channel. As the cell sizes are the same for both MCF-7 and MCF-10A in DPD simulation, this indicates that MCF-10A is less deformable than MCF-7. This difference in entry times can be attributed to malignant MCF-7 cells being about 4–5 times less stiff compared to benign MCF-10A cells [38]. However, the linear portion of the graph in the transit region for MCF-10A has similar gradients as compared with MCF-7. Hence, this may suggest that the transit velocities of the two types of cells may be similar. Our findings from DPD simulations agrees with Hou et al.'s [4] experimental observation.

5. Conclusion

This paper presented a DPD model for simulating the movement and deformability of single cells when entering micro-channels with confined structures. The entry processes of MCF-10A and MCF-7 cells are comparatively investigated. We conclude that the presented DPD method with WLC spring to model cells is effective in simulating cell entry into micro-channels. Both the cell displacement time-plots and the dynamical behavior of the cell entry are shown to match quantitatively throughout the cell entry, subjected to experimental uncertainties and model assumptions. Furthermore, the model allowed us to confirm that MCF-10A cells have longer entry time than MCF-7 cells of similar sizes because MCF-10A cells are stiffer than MCF-7 cells. These quantitative agreements show the usefulness and effectiveness of our models in interpreting cellular experiments.

Nevertheless, there are several limitations of this model. Firstly, the model assumed that the cell's volume keeps nearly constant during deformation. The incompressibility of the cell in simulation leads to

more time required for the cell to enter the micro-channel, and stretch longer in the micro-channel than in experiment. Secondly, flow in the micro-channel is driven by a differential pressure in experiment, and we use a body force with periodic boundary condition in x direction instead of differential pressure in DPD simulation. As such, the simulation domain is not exactly the same as experiment setup especially when computational domain is small. Thirdly, the nucleus is known to be able to influence the entry time, while the current models is not able to consider cell nucleus. Fortunately, It is convenient to add cell nucleus in the DPD model by just adding a new network formed from DPD particles and spring chain, provided that parameters of cell nucleus are known.

Conflict of interest

All authors disclose that there was no conflict of interest regarding this work.

Acknowledgement

This work was supported by National Natural Science Foundation of China grants 11502272 to Shiliang Feng; and the K.C. Wong Magana Fund through Ningbo University to Lüwen Zhou.

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